Songjie Jia

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Research interests: Self-adaptive structures, soft robotics, mechanics-based simulation and imitation learning.

Education

Michigan Technological University Jan. 2025–Present
PhD Candidate, Civil Engineering *Houghton, MI*
- Dissertation proposal: *"Robotic Adaptive Structures: Bridging Robots and Civil Structures."*
- Selected coursework: Machine Learning, Control Systems; **GPA: 3.78**.

University of Michigan Aug. 2021–Sept. 2022
Master, Civil Engineering *Ann Arbor, MI*
- Selected coursework: Nonlinear Analysis, Finite Element Analysis, Plastic Analysis; **GPA: 3.587**.

Nanjing Tech University Sept. 2016–June 2020
Bachelor, Civil Engineering *Nanjing, China*
- Selected coursework: Structural Analysis, Mechanics of Materials, Reinforced concrete; **GPA: 85/100**.

Journal Publications

- S. Jia and Y. Zhu, "Cable Pneumatic Robot: Fabrication and Simulation," *Soft Robotics*, Apr. 2026. doi: 10.1177/21695172261442061.
- S. Jia and Y. Zhu, "BLOC: Blowable Origami Cube for Modular, Deployable, and Reusable Structural Systems," in submission cycle.

Awards

- **2nd Place**, ICC Student Poster Competition, Michigan Tech, 10/2025;
- **Second-class Scholarship**, Nanjing Tech University, 4/2020;
- **Freshman Young Eagle Scholarship**, Nanjing Tech University, 09/2016

Research Experience and Projects

Cable Pneumatic Robot Mar. 2025–Oct. 2025
- Created an integrated fabrication and simulation framework for cable pneumatic soft robots that combine large pneumatic shape morphing with cable-driven controllability.
- Developed a lumped-parameter simulation model that reduces computational complexity relative to conventional finite element methods while capturing robot deformation behavior.

Modular Self-reconfigurable Structures Nov. 2025–Sept. 2026
- Fabricated pressure-controlled modular adaptive structures with rigid skeletons and inflatable actuators, including self-reconfigurable frame and arc-bridge configurations.
- Developed simulation of reconfigurable structures that transition between compact and load-bearing states.

Imitation Learning for Robotic Deployment of Origami-Inspired Structures May 2026–Present
- Exploring imitation learning from human demonstrations to teach a robotic manipulator to unfold and position foldable adaptive structures.

Conference

- Songjie Jia, 2026, Cable Pneumatic Robot: Fabrication and Simulation, Society of Engineering Science, Oct 11-14 2026, Purdue University, West Lafayette, Indiana. (Poster)

Technical Skills

- **Simulation/Programming:** MATLAB, Python, Abaqus, COMSOL.
- **Fabrication/Experiment:** SolidWorks, CAD, 3D printing, laser cutting.
- **Robotics/Hardware:** Arduino, sensors, microcontrollers, Linux/Ubuntu, ROS 2.

Industry Experience

Structural Designer Sept. 2022–Oct. 2023
Architectural Design & Research Institute of Tongji University *Yangpu, Shanghai, China*
- Contributed to the structural design of a large-scale mixed-use complex, with independent responsibility for the structural design of two office buildings.

Services and Additional Experience

- **President, Chinese Students and Scholars Association**, Michigan Tech Aug. 2026–Present